Elements of Statistical Learning: Lecture 2. Multiple Regression and Variable Selection

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Multivariate Linear Model

Now suppose we have p explanatory variables.

Call them $\mathbf{X} = (X_1, \ldots, X_p)$.

We have the values of $\mathbf{X}$ on n subjects.

The outcome (response) for the ith subject is Y_i (univariate).

And exploratory data analysis (such as plots of X_j against Y) indicate that a linear model is appropriate.

So, could write

$$Y_i = X_{i,1}\beta_1 + X_{i,2}\beta_2 + \cdots X_{i,p}\beta_p + \epsilon_i,$$

for $i = 1, \ldots, n$. I.e. n equations, so clearer to write in matrix form.

Matrix formulation of multivariate linear model

Let $Y = (Y_1, \ldots, Y_n)^T$ and $\epsilon = (\epsilon_1, \ldots, \epsilon_n)$.

Let $\beta = (\beta_1, \ldots, \beta_p)$ and $X = (X_{i,j})$ the $n \times p$ matrix containing elements $X_{i,j}$ values of $\mathbf{X}$ on the ith individual and jth variable.

Then, we can write the multivariate linear model as

$$Y = X\beta + \epsilon. \tag{1}$$

Here ϵ is a n-vector of IID random variables with mean zero and variance σ^2 as before. We assume Gaussianity.

Residuals, sum of squares, least squares

The residual, as a function of β, can be defined as a vector e

$$e = Y - X\beta.$$

The residual sum of squares as

$$\mathrm{RSS}(\beta) = e^T e.$$

The least squares estimator is

$$\hat{\beta} = \mathrm{argmin}_\beta \, RSS(\beta).$$

And the (empirical) residual is usually written as

$$e = Y - X\hat{\beta}. \tag{2}$$

Example: simple linear regression

Here, $Y = (Y_1, \ldots, Y_n)$, $\epsilon = (\epsilon_1, \ldots, \epsilon_n)$ and

$\beta = (a, b)^T$ and

$$X = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}.$$

For model $Y_i = a + bx_i + cx_i^2 + \epsilon_i$ then Y, ϵ are the same, but $\beta = (a, b, c)^T$ and

$$X = \begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 \end{pmatrix}$$

.

Lemma: vector and matrix differentiation

Lemma 1. Let $u = \beta^T v = \sum_{i=1}^{p} \beta_i v_i$, where v is a p-vector.

Then

$$\frac{\partial u}{\partial \beta_j} = \sum_{i=1}^{p} v_i \frac{\partial \beta_i}{\partial \beta_j} = v_j,$$

so $\frac{\partial u}{\partial \beta} = v$.

Lemma 2. Let the quadratic form $w = \beta^T A \beta$ for some symmetric matrix A. We can write

$$w = \sum_{i=1}^{p} \beta_i \sum_{j=1}^{p} A_{i,j} \beta_j,$$

then $\frac{\partial w}{\partial \beta} = 2A\beta$ (see exercises).

Least squares estimation

Write RSS as

$$\begin{aligned} \text{RSS}(\beta) &= (Y - X\beta)^T (Y - X\beta) \\ &= Y^T Y - (X\beta)^T Y - Y^T X\beta + (X\beta)^T X\beta \\ &= Y^T Y - 2\beta^T X^T Y + \beta^T X^T X\beta. \end{aligned}$$

Differentiating with respect to β

$$\frac{\partial \text{RSS}(\beta)}{\partial \beta} = -2X^T Y + 2X^T X\beta \tag{3}$$

Setting equal to zero and solving for β gives

$$X^T Y = X^T X\hat{\beta} \implies \hat{\beta} = (X^T X)^{-1} X^T Y$$

$X^T X$ is invertible if X is of full rank.

Orthogonality of residual vector to columns of X

First, note that

$$\frac{\partial^2 \text{RSS}(\beta)}{\partial \beta \, \partial \beta^T} = 2X^T X,$$

which is positive definite (since $X^T X$ is invertible) and hence, we have a minimum at this point.

Second, note that, in solving the equations $\frac{\partial \text{RSS}(\beta)}{\partial \beta} = 0$ on the previous slide we can write

$$X^T (Y - X\hat{\beta}) = 0,$$

i.e. we've substituted the solving value of $\beta = \hat{\beta}$. The $(Y - X\hat{\beta})$ is the residual of the least squares fit. So, the residual vector of the fit is orthogonal to the columns of X.

Hat matrix

Define $H = X(X^T X)^{-1} X^T$ to be the hat matrix.

Then the fitted values of the model are

$$\hat{Y} = X\hat{\beta} = HY,$$

so that hat matrix puts the hat on Y.

Note that:

1. Parameter estimates $\hat{\beta}$ are linear combinations of the Y, and
2. fitted values, $\hat{Y}$, are also linear combinations of the Y.

Inference about β: Expectation, variance of Y

First, remember $\mathbb{E}(Y) = X\beta$, from (1).

The $p \times p$ variance matrix of random p-vector U is defined as

$$\Sigma = \text{var}(U) = \mathbb{E}\left[\{U - \mathbb{E}(U)\}\{U - \mathbb{E}(U)\}^T\right] \tag{4}$$

so $\text{cov}(U_i, U_j) = \Sigma_{i,j}$

Hence,

$$\begin{aligned} \text{var}(Y) &= \mathbb{E}\left[\{Y - \mathbb{E}(Y)\}\{Y - \mathbb{E}(Y)\}^T\right] \\ &= \mathbb{E}(\epsilon\epsilon^T) = \Sigma_\epsilon = \sigma^2 I_n, \end{aligned}$$

where I_n is the n-dimensional identity matrix.Also:

$$\sigma^2 I_n = \mathbb{E}(YY^T) - \mathbb{E}(Y)\mathbb{E}(Y^T) \implies \mathbb{E}(YY^T) = \sigma^2 I_n + X\beta\beta^T X^T.$$

Inference about β: Expectation, variance of β

Expectation

$$\begin{aligned}\mathbb{E}(\hat{\beta}) &= \mathbb{E}\left\{(X^T X)^{-1} X^T Y\right\} \\ &= (X^T X)^{-1} X^T \mathbb{E}(Y) = (X^T X)^{-1} X^T X \beta = \beta.\end{aligned}$$

So, least-squares estimator is unbiased.What about variance?

$$\begin{aligned}\text{var}(\hat{\beta}) &= \mathbb{E}\left[\{\hat{\beta} - \mathbb{E}(\hat{\beta})\}\{\hat{\beta} - \mathbb{E}(\hat{\beta})\}^T\right] \\ &= \mathbb{E}(\hat{\beta}\hat{\beta}^T) - \mathbb{E}(\hat{\beta})\mathbb{E}(\hat{\beta}^T) \\ &= \mathbb{E}\left[(X^T X)^{-1} X^T Y Y^T X (X^T X)^{-1}\right] - \beta\beta^T \\ &= (X^T X)^{-1} X^T \mathbb{E}(YY^T) X (X^T X)^{-1} - \beta\beta^T \\ &= (X^T X)^{-1} X^T (\sigma^2 I_n + X\beta\beta^T X^T) X (X^T X)^{-1} - \beta\beta^T \\ &= \sigma^2 (X^T X)^{-1}\end{aligned}$$

Explanation about var($\hat{\beta}$)

Here var($\hat{\beta}$) is a $p \times p$ covariance matrix.

The covariance between $\hat{\beta}_i$ and $\hat{\beta}_j$ is $\left\{\sigma^2(X^T X)^{-1}\right\}_{i,j}$.

The parameter variance depends on X, i.e. the design, which can sometimes be controlled with proper planning.

For example, if X is an orthogonal matrix then $X^T X = I_p$ and var $\hat{\beta}$ is a diagonal matrix, so $\hat{\beta}_i$ and $\hat{\beta}_j$ are uncorrelated for $i \neq j$.

This is practically useful because it means that different variables can be considered individually, without reference to the others.

Inference about β: Distribution of β

If the $\{\epsilon_i\}$ are normally distributed $N(0, \sigma^2)$ then, since $\hat{\beta}$ is a linear combination of normals, then it must be normal itself and we must have

$$\hat{\beta} \sim N_p\{\beta, \sigma^2(X^T X)^{-1}\}.$$

(And if X is orthogonal, then $\hat{\beta}_i$ and $\hat{\beta}_j$ are independent, for $i \neq j$.)

[Because uncorrelated and Gaussian $\implies$ independent]

Estimating σ^2

Let v_j be the jth diagonal element of $(X^T X)^{-1}$. Then $\text{var}(\hat{\beta}_j) = \sigma^2 v_j$. Then the standardised coefficient or Z-score is

$$z_j = \frac{\hat{\beta}_j}{\hat{\sigma}\sqrt{v_j}}, \tag{5}$$

where we typically estimate σ^2 by

$$\hat{\sigma}^2 = (n-p)^{-1} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2,$$

and $\mathbb{E}\hat{\sigma}^2 = \sigma^2$.

Under the model assumptions and the null hypothesis $H_0 : \beta_j = 0$, the quantity z_j is distributed as a Student's t-distribution on $n-p$ degrees of freedom (and usually approximated by $N(0,1)$ for n large, and larger than p).

p or $p+1$?

When we add a mean or intercept parameter into the model, e.g. β_0, as we will do later, this can be included in the design matrix X as a column of 1s as on Slide 5.

This increases the number of parameters from p to $p+1$, and then the $n-p$ quantities on the previous slide become $n-p-1$.

Of course, for even moderately large n the difference between $1/(n-p)$ and $1/(n-p-1)$ is not large, unless, of course, p is large too.

Inference about β: terminology

We are interested in whether β_j is zero or not.We calculate z_j and compare this to $N(0,1)$ (or appropriate t).

The p-value is the probability of getting a more extreme value than z_j, if assuming it comes from a $N(0,1)$ distribution.

We are indifferent to positive or negative values so, we compute

$$p_{z_j} = \mathbb{P}(|Z| > z_j) = \mathbb{P}(Z < -z_j) + \mathbb{P}(Z > z_j) = 2\mathbb{P}(Z < -z_j) = 2\Phi(-z_j),$$

where $Z \sim N(0,1)$ and its distribution function is Φ.

Usually say if $p_{z_j} < 0.01$ then strong evidence to reject H_0, else $p_{z_j} < 0.05$ some evidence to reject H_0, otherwise have little or no evidence to reject H_0 (but this does NOT mean accept H_0).

Bayesian methods give distribution for z_j, might be more useful.

Dropping/Including Groups of Variables

Consider two models: model M0 and model M1.

Model M1 is the larger model with $p_1 + 1$ variables.

Model M0 is smaller, with $p_0 + 1$, all contained in M1 (nested).

If we go from the larger to the smaller, we're constraining $p_1 - p_0$ parameters to be zero.

Calculate

$$F = \frac{(\text{RSS}_0 - \text{RSS}_1)/(p_1 - p_0)}{\text{RSS}_1/(n - p_1 - 1)},$$

Note, $\text{RSS}_1 \leq \text{RSS}_0$ as M1 has more parameters (better fit).

Inference about models

Under the model assumptions and that M0 (smaller) is true, it can be shown that $F \sim F_{p_1-p_0,n-p_1-1}$ distribution, which gives us a way of assessing which model to prefer as our working assumption going forward.

Child Data Example

The child data has two other variables: `age` and `sex`.

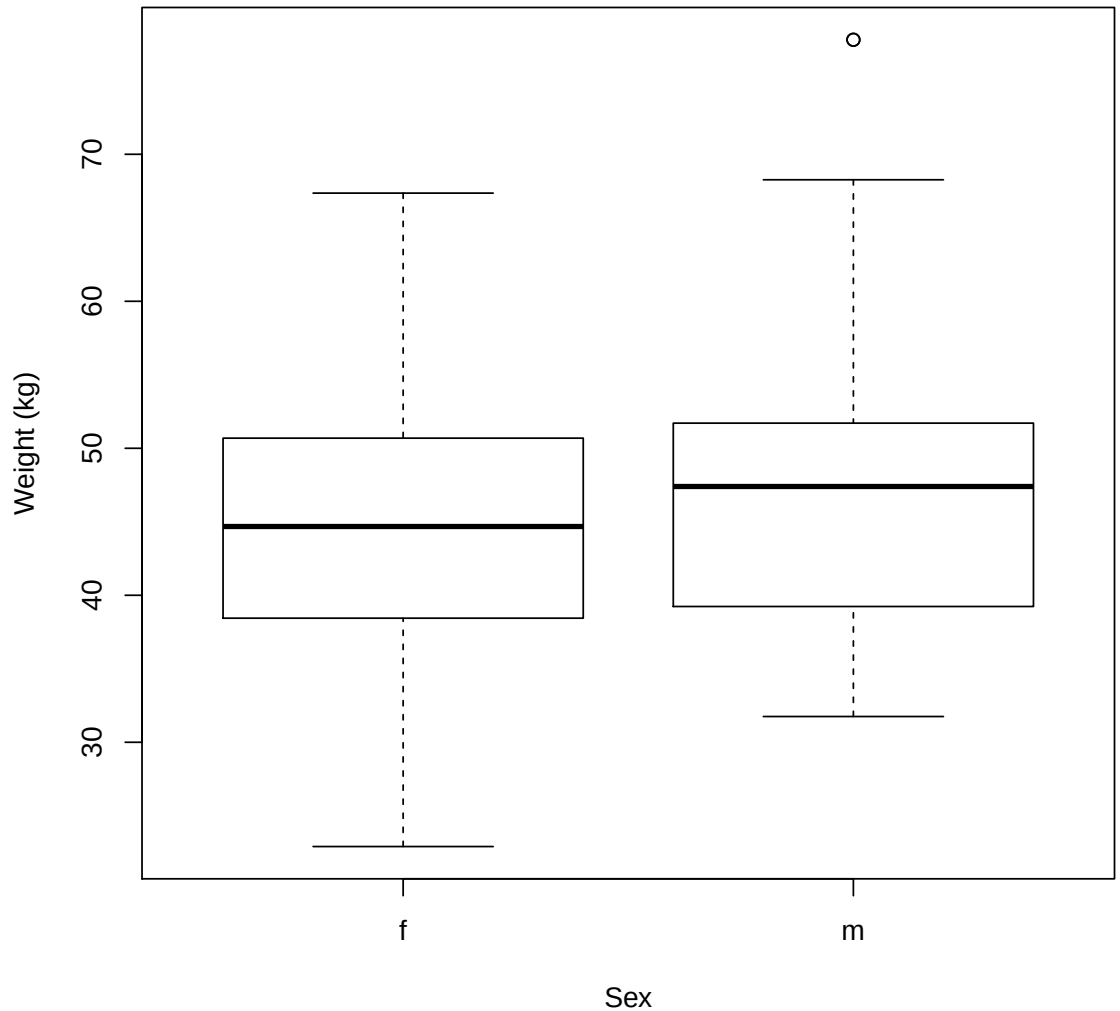
All the variables are contained in the `childhwDF` data frame.

Let's look at some exploratory plots.

The `R` function `split`, splits a variable into (two) groups depending on a (dichotomous) factor.

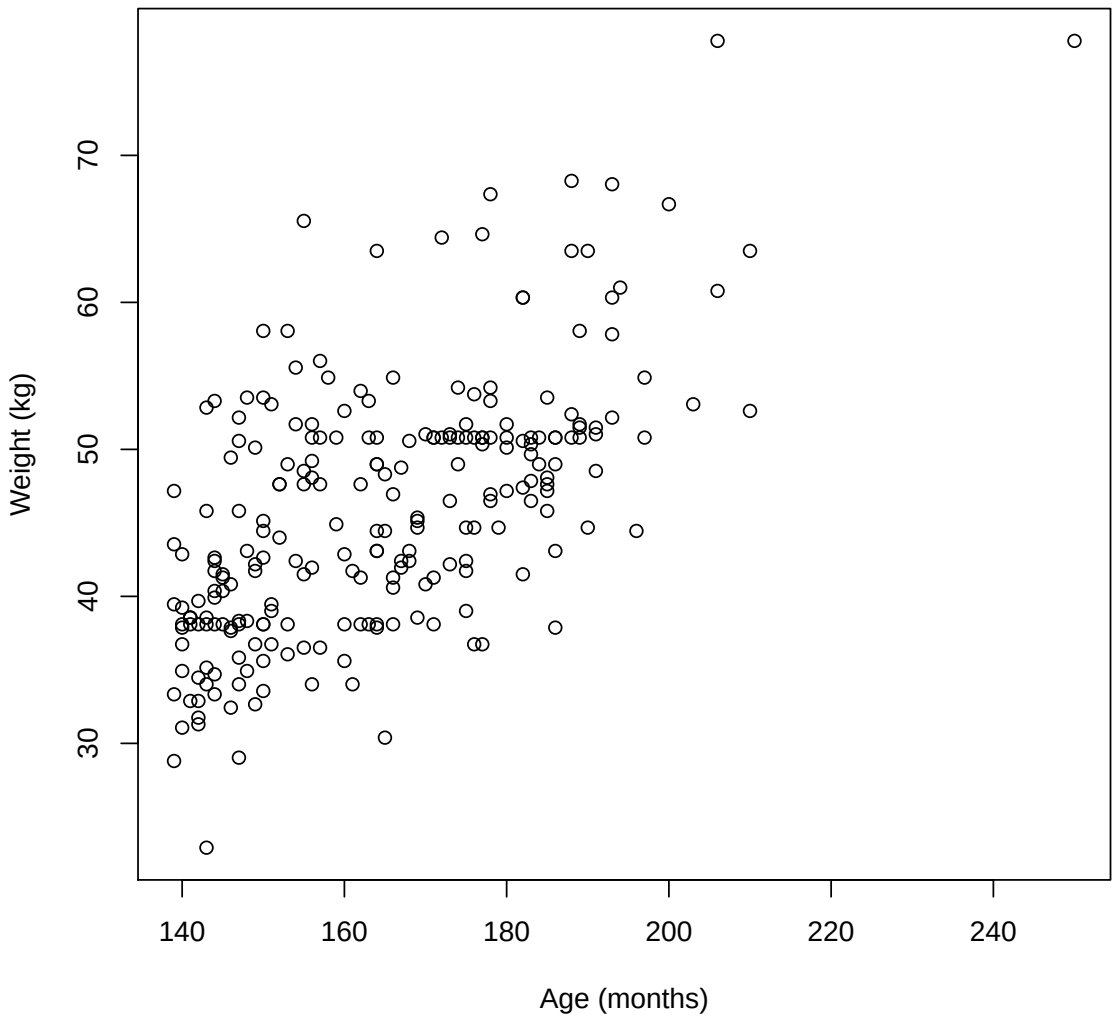
Weight split by sex

```
attach(childhwDF)
boxplot(split(x=weight, f=sex))
```



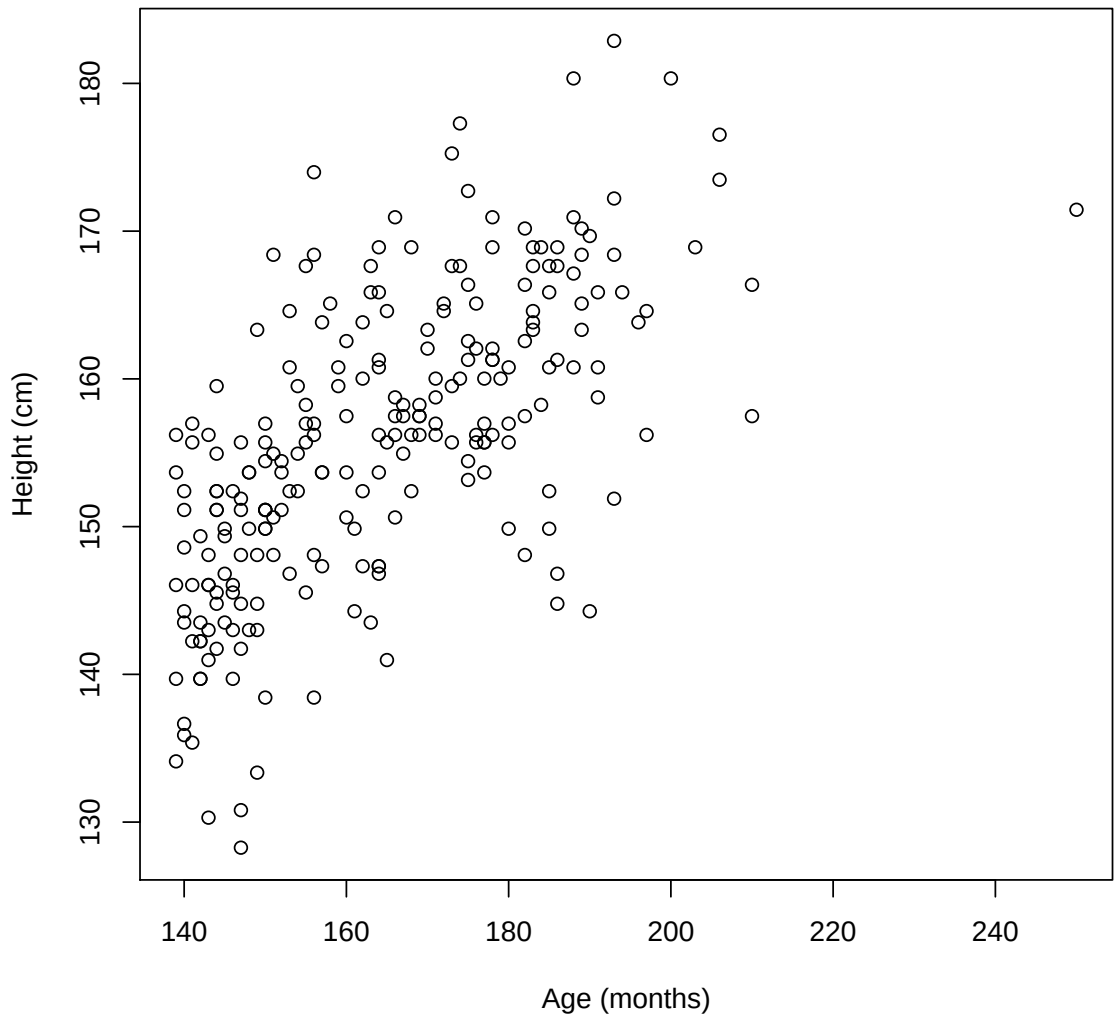
Weight against age

```
plot(age, weight, xlab="Age (months)", ylab="Weight (kg)")
```



Height against age

```
plot(age, height, xlab="Age (months)", ylab="Height (cm)")
```



F-test on the three variables

```
#
# Fit full model with all three variables and intercept
#
fit1 <- lm( weight~ height+sex+age, data=childhwDF)
#
# Now just fit intercept
#
fit0 <- lm( weight~ 1, data=childhwDF)
#
# Calculate RSS for both models
#
rss1 <- sum(resid(fit1)^2)
rss0 <- sum(resid(fit0)^2)
```

F-test on the three variables (continued)

```
#
# Calculate F statistic
#
((rss0 - rss1)/(4-1))/(rss1/(237-4-1))
[1] 131.9756
#
# 99% quantile of appropriate F
#
qf(0.99, df1=3, df2=237-4-1)
[1] 3.867119
```

The null hypothesis would be $\beta_{\text{sex}} = \beta_{\text{age}} = \beta_{\text{height}} = 0$

Now 132.0 $\gg$ 3.87, so have strong evidence to reject the null hypothesis.

Stepwise regression: backward deletion

`summary(fit1)`

Start with full model and look at coefficients and s.e.'s

	Estimate	Std. Error	t value	$\mathbb{P}(>\|t\|)$
(Intercept)	-58.15	5.56	-10.45	$< 2 \times 10^{-16}$
`height`	0.55	0.048	11.62	$< 2 \times 10^{-16}$
`sex-m`	-0.15	0.73	-0.21	0.834
age	0.11	0.025	4.25	3.1×10^{-5}

The sex variable is not significant, so we drop it from the regression:

Stepwise regression: backward deletion

```
fit2 <- update(fit1, . ~ . -sex)
summary(fit2)
```

	Estimate	Std. Error	t value	$\mathbb{P}(>\|t\|)$
(Intercept)	-57.98	5.49	-10.56	$< 2 \times 10^{-16}$
`height`	0.55	0.046	12.01	$< 2 \times 10^{-16}$
`age`	0.11	0.025	4.36	1.95×10^{-5}

Backward-stepwise selection: starts with the full model and sequentially delete the predictor that has least input on the fit.

I stop when all variables are statistically significant at the 5% level.

Forward-stepwise selection

Start with intercept in model
`fitInt <- lm(weight ~ 1, data=childhwDF)`

Sequentially add the term that most improves the fit.

If you have p variables to add, add them one by one and see which one improves the fit the most.

Stop when the fit stops improving, or when the introduced variable is is not statistically significant (i.e. the coefficient in the model is not statistically significant).

Summary

This lecture introduced the following concepts:

The Multivariate Linear Model

Vector and matrix differentiation

Computing Least Square Estimates for the multivariate linear model

Orthogonality between the residual vector and the columns of X

Hat matrix

Expectation and Variance matrix of $\hat{\beta}$

Distribution of $\hat{\beta}$ and estimation of σ^2.

p-values

Variable Selection groups, backward deletion and forward stepwise

Child Data example